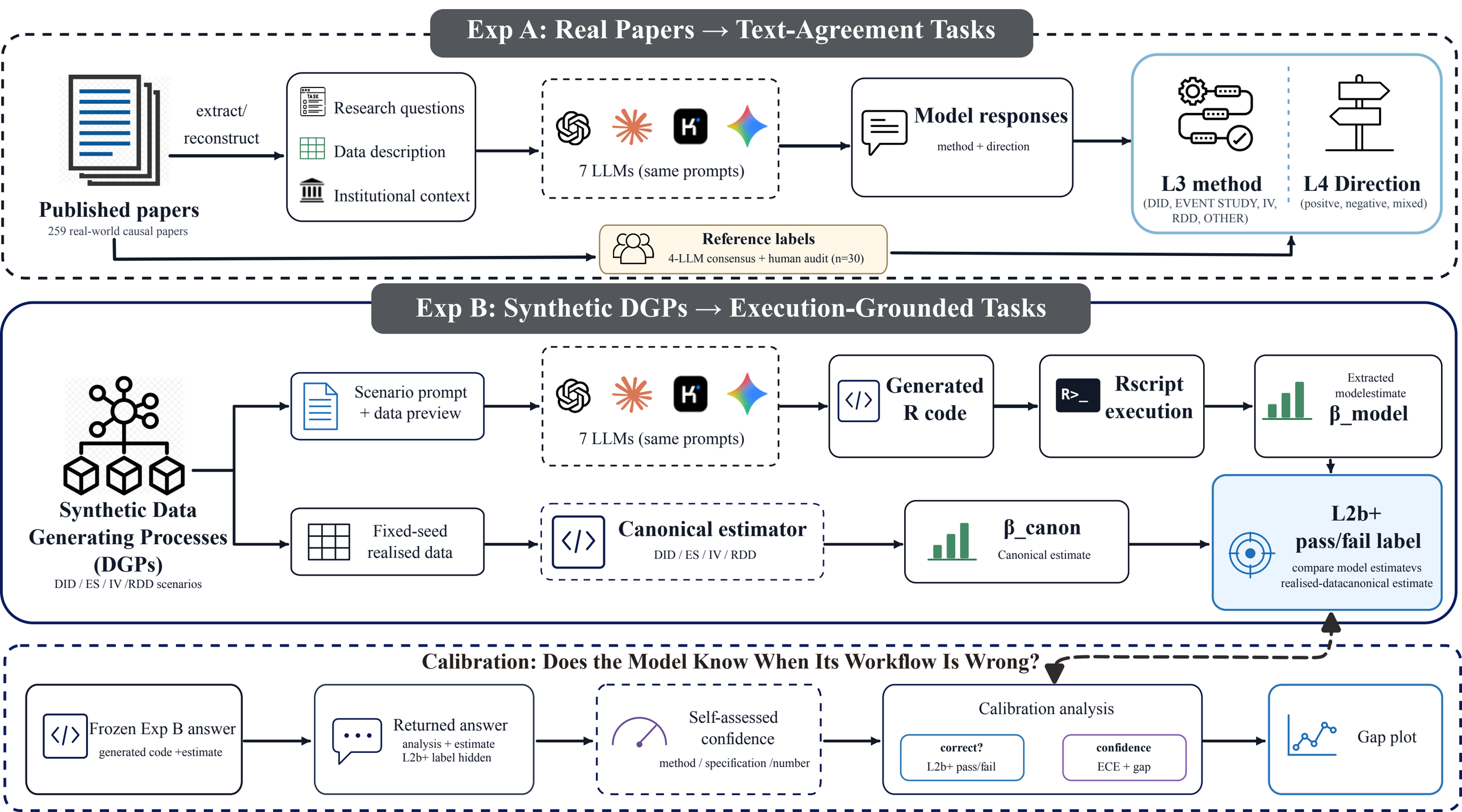

Exp A: Real Papers → Text-Agreement Tasks
Published papers
259 real-world causal papers
extract/
reconstruct
Research questions
Data description
Institutional context
7 LLMs (same prompts)
Model responses
method + direction
L3 method
(DID, EVENT STUDY, IV, RDD, OTHER)
L4 Direction
(positve, negative, mixed)
Reference labels
4-LLM consensus + human audit (n=30)
Exp B: Synthetic DGPs → Execution-Grounded Tasks
Synthetic Data Generating Processes (DGPs)
DID / ES / IV /RDD scenarios
Scenario prompt + data preview
7 LLMs (same prompts)
Generated R code
Rscript execution
Extracted modelestimate
β_model
Fixed-seed realised data
Canonical estimator
DID / ES / IV / RDD
β_canon
Canonical estimate
L2b+ pass/fail label
compare model estimatevs realised-datacanonical estimate
Calibration: Does the Model Know When Its Workflow Is Wrong?
Frozen Exp B answer
generated code +estimate
Returned answer
analysis + estimate
L2b+ label hidden
Self-assessed confidence
method / specification /number
Calibration analysis
correct?
L2b+ pass/fail
confidence
ECE + gap
Gap plot